Synthese (2026) 208:224
https://doi.org/10.1007/s11229-026-05814-4

ORIGINAL RESEARCH

What philosophy can learn from AI: methodological insights from deep learning and Chinese philosophy

Paul J. D'Ambrosio[1,2]

Received: 18 June 2026 / Accepted: 1 September 2026
© The Author(s), under exclusive licence to Springer Nature B.V. 2026

Abstract

As artificial intelligence exerts growing influence across society, academic philosophers are increasingly called upon to address questions concerning ethics, alignment, agency, and meaning. Yet recent developments in AI can also invite reflection on certain prominent philosophical methodologies as well. Appreciating the success of deep learning systems raises questions about the degree to which effective understanding depends upon explicit definitions, conceptual clarification, and linear forms of reasoning. It also invites us to seriously reconsider the value of examples, analogies, patterns, tacit knowledge, and context-sensitive responsiveness in understanding and interacting in the world. This paper develops a limited structural comparison between selected methodological tendencies in contemporary Anglophone academic philosophy and supervised classical machine learning. For example, although the latter is statistical and driven by data, both tend to rely comparatively heavily upon predefined categories, predetermined structures, human-selected features, and relatively transparent forms of reasoning. By contrast, deep learning and selected approaches in early Confucian and Daoist philosophies exhibit noteworthy similarities in their reliance upon examples, cultivation, and iterative engagement with experience. While not equivalent, both do illustrate how productive patterns of response can be developed through repeated encounters with particulars, while relying less on pregiven specified conditions. We therefore argue that the success of deep learning invites renewed reflection on certain academic assumptions that effective forms of understanding and competence must depend heavily upon explicit conceptual articulation. In sum, the paper suggests that recent developments in AI provide an opportunity to reconsider the methodological scope of philosophical inquiry itself. We may also make space for philosophical approaches where inquiry begins with a variety of provisional models, examples, doctrines, or patterns, and then tests them, and iteratively revises them, in actual cases, according to specific experiences and contexts.

Extended author information available on the last page of the article

Published online: 09 November 2026

Keywords Artificial intelligence · Deep learning · Confucianism · Daoism · Iterative learning · Philosophical methodology

1 Introduction

As AI systems exert growing influence across society, academic philosophy professors have a renewed public significance. Some of the most influential contributions to AI, including ideas which have inspired technological advances, have come from individuals with philosophical training.[1] Yet professional philosophers are more often called upon to address ethical concerns, including various biases, unfair practices, and issues of accountability in the development and use of AI systems. Scholars hailing from the tradition of Plato and Aristotle are also expected to be part of the more forward-looking discussions of AI alignment, value learning, and human-in-the-loop models. Most recently, the emergence of large language models (LLMs) such as ChatGPT has given rise to deeper existential challenges: If AI can perform many of the tasks we long considered uniquely human, what does it mean to be human? Philosophy professors are then also tasked with reflecting on how we might reconstruct meaning if AI makes many jobs obsolete, and, looking even further ahead, how purpose, meaning, and identity might be constructed in a "solved" world. In sum, the rapid development of AI has brought forth ethical, political, and existential challenges which have prompted renewed interest in philosophical inquiry.

As the remarkable achievements in AI are causing us to rethink everything from driving and banking to writing and teaching, we might also pause to reconsider our philosophical approaches before haphazardly assuming they are already well suited to address the abovementioned problems. In this paper we will argue that recent developments in AI invite reflection not only on technology, but on philosophy itself. In particular, the success of deep learning systems raises questions about the degree to which understanding must depend upon explicit definitions, conceptual clarification, and formal reasoning. At the same time, we open space to see how examples, analogies, patterns, tacit knowledge, and context-sensitive responsiveness may also constitute important forms of understanding. This, we will argue, is true not only for the content of understanding, or the content of philosophical inquiry, but also in terms of method.[2]

In other words, we will argue that in addition to AI learning from philosophy, philosophy stands to learn from AI. Specifically, we suggest that certain forms of

[1] Examples include Nick Bostrom, whose work has significantly influenced contemporary AI discussions; Murray Shanahan, who works at DeepMind and has written extensively on artificial intelligence and philosophy; and DeepMind co-founder Mustafa Suleyman, who studied philosophy before entering the technology sector.

[2] Our argument is that deep learning can provide a strong counterexample to the assumption that competence must be preceded by exhaustive pregiven specified conditions. However, this is not to say that deep learning somehow proves that Confucian and Daoist philosophy are valuable. Instead, the selected approaches from Confucian and Daoist traditions are used to demonstrate that we in fact already have resources for considering how philosophical reflection does not need to depend heavily upon explicit conceptual articulation.

philosophical reflection operate in ways structurally analogous to classical machine learning, while others bear noteworthy similarities to deep learning. This comparison intersects well with longstanding philosophical discussions concerning tacit knowledge,[3] embodied competence, and the limits of rule-based accounts of intelligence, particularly those associated with Michael Polanyi and Hubert Dreyfus.[4] In addition to a focus on developing exact definitions of concepts—a method that parallels certain forms of classical machine learning—philosophical reflection can also involve exploring examples, models, analogies, and other pattern-based practices. And it may rely heavily on interactions and on developing a dynamic sensitivity to particulars. This type of philosophizing leaves much of the detailed explanation out of the picture; either because it can be developed in other spheres, or simply because it is not required given the particular contexts involved.

Compared with much contemporary Anglophone academic philosophy, this would constitute a significant methodological shift. If our concern was, for example, the nature of "devotion," we would be asking what it looks and feels like, how it is related to other phenomena, and associated concrete questions, rather than clarifying or asking "what exactly *is the essence* of devotion" and, "what universally unites various types of devotion." We could appreciate how identifying multiple examples of devotion could teach us a lot about it—how it is expressed, cultivated, and challenged—without requiring specific parameters of "what devotion is." Accordingly, these varied cases would allow philosophers to formulate provisional accounts of devotion, and to test them in real-world settings. They could look at practical consequences, and revise their understanding iteratively. Standard methods, such as hypothetical cases and abstract arguments would remain important, but they would be one strategy among others, or perhaps even one component of a broader process.

This paper is structured in the following way: in the next section we will give a general discussion of some dominant tendencies in contemporary Anglophone academic philosophy. The third section will provide an equally general overview of supervised classic machine learning. We will then trace some similarities between these philosophical and computational approaches in section four. Then in section five we will look at the structure of deep learning algorithms, and in the sixth section we will introduce selected methodological orientations found in early Confucian and Daoist texts. In section seven we explore methodological parallels between deep learning and early Confucian and Daoist philosophies. Finally, in the conclusion we consider what philosophy might learn from developments in AI. Given the breadth

[3] Tacit knowledge is an established framework which can help introduce the gap between effective performance and explicit articulation. However, most of our argument will place emphasis on iterative feedback, correction, and revision. So while we do think there is space for developing a rich discussion between the arguments given here and theories of tacit knowledge, we will not delve into this theme—it is simply too broad for the scope of this paper.

[4] The broad scope of this project does not allow for a more detailed discussion of these discussions or of Polanyi and Dreyfus, and the larger discourses involved. That might be the subject of future work, which would build off the foundations laid here.

and complexity of the topics discussed here,[5] all overviews and comparisons are necessarily general, and important exceptions exist in each case.

2 Orientations in contemporary Anglophone academic philosophy

This section will outline major orientations in contemporary Anglophone academic philosophy (hereafter CAAP). No such discussion could ever be comprehensive, and reasonable disagreement can be expressed regarding any of the points made below. The three main characteristics discussed here are meant to identify some of the prevailing methodological tendencies that remain common within academic philosophy conducted in English today. There are important exceptions to each of these tendencies, and the purpose is not to exhaustively define CAAP. Rather, the goal is to identify a set of recurring methodological commitments that can be placed into a productive dialogue with AI technologies and, later, with early Confucian and Daoist approaches to philosophical reflection. We thereby use CAAP as a heuristic designation which indicates selected influential methodological tendencies found in certain philosophical discourses. CAAP does not name a unified tradition nor is it representative of some transhistorical practice.

Before beginning it should also be emphasized that the discussion below concerns broad tendencies and not fixed or universal characteristics. Many important exceptions exist within the Anglophone tradition itself. For example, Wittgenstein's notion of family resemblance challenges the assumption that meaningful concepts must always be defined through necessary and sufficient conditions or narrow definitions (Wittgenstein, 2009). Likewise, phenomenological approaches often emphasize lived experience over conceptual analysis (Moran, 2000). Nevertheless, the three tendencies discussed here—namely, conceptual clarification, rational argument, and emphasis on essentials—remain influential orientations within much contemporary Anglophone philosophy.

Philosophy has always been notoriously difficult to define, especially for philosophers. Nevertheless, some informative descriptions can be found. The American Philosophical Association website, for example, provides the following account for undergraduates interested in the major:

> It is a reasoned pursuit of fundamental truths, a quest for understanding, a study of principles of conduct. It seeks to establish standards of evidence, to provide rational methods of resolving conflicts, and to create techniques for evaluating ideas and arguments. […]
>
> As the systematic study of ideas and issues, […] philosophical studies often focus on the meaning of an idea and on its basis, coherence, and relations to other ideas. Consider, for instance, *democracy*. What is it? What justifies it as

[5] The content of this paper argues for a more serious appreciation of examples, analogies and patterns in philosophical inquiry. It also engages, to some degree, in this type of methodology—which is necessarily limiting in some ways.

> a system of government? Can a democracy allow the people to vote away their own rights? And how is it related to political liberty? (American Philosophical Association, n.d.).

Already in this general discussion we find the three methodological orientations that will be important for the present discussion. First, there is a strong emphasis on reason and rational arguments. Second, we see a concern for conceptual clarification, including an analysis of concepts themselves and their relations to each other. Third, there is a tendency to distinguish what is essential from what is accidental, peripheral, or non-essential. These tendencies are not unique to philosophy, nor are they present in all philosophical traditions. They are, however, influential in many areas of CAAP.

Reliance on reason and rational arguments constitutes a major aspect of CAAP. One attraction of reason-centered approaches is that they appear capable of transcending historical, cultural, and personal differences. Rational argument promises a mode of communication which addresses individuals as rational agents rather than members of particular communities or traditions.

A paradigmatic example is found in Plato's *Meno*. Although the portrayed Socrates is ostensibly demonstrating that learning is recollection,[6] the dialogue can equally be interpreted as demonstrating the power of step-by-step rational argument (Plato, *Meno*, 1997, pp. 81a–86c). By and large, contemporary academic philosophy functions similarly—which does not, however, necessarily indicate historical continuity. Articles, books, and conference presentations frequently attempt to construct arguments whose conclusions should be persuasive to anyone capable of following the reasoning. This remains true even in areas such as moral psychology, philosophy of emotion, and care ethics, where lived experience and affective dimensions are central topics of inquiry. Here too reason-based arguments and rational descriptions remain dominant. (The *Meno* is thus on example of such type inquiry, and whether it is directly linked to today's academic discourse remains outside the scope of this project.)

A strong reliance on reason is not unique to philosophy. Many academic disciplines prioritize rational arguments, especially those that aspire to scientific rigor. CAAP further distinguishes itself insofar as it is concerned with clarifying concepts. As suggested in the APA's description, philosophy frequently asks questions such as "What is democracy?" The expectation is not merely to provide a description, but to determine what democracy is, how its various forms relate to one another, and how it differs from alternative forms of political organization. Truly, refined conceptual distinctions reign among the central achievements of CAAP inquiry.

For example, in the discourse on moral luck, dozens of articles have been written not just on different types of luck,[7] but on further subsets of a particular type, or on the details of the relation to other concepts. Questions then arise regarding whether resultant luck affects the scope or degree of moral responsibility, and whether the

[6] Socrates' proof is demonstrated in proposing a mathematical problem to one of Meno's slaves, a person with no formal education.

[7] Much of the contemporary literature takes Thomas Nagel's original categories—namely, resultant, constitutive, circumstantial, and causal luck—as a starting point (Nagel, 1979).

responsibility involved is internal or external, and how such distinctions interact with broader theoretical commitments (Rolffs, 2023). These discussions are often rigorous and illuminating. Nevertheless, critics will note that their increasing specialization can create a noticeable distance between philosophical discourse and the concerns of those outside the field.

The search for conceptual clarity is closely related to a third tendency: the pursuit of what is fundamental, essential, or non-contingent. One common strategy here is to identify what different instances of a phenomenon share while setting aside features that appear contingent or accidental. Asking "What is democracy?" is therefore often understood as a search for those characteristics that make something democratic. Philosophical analysis attempts to isolate those characteristics and distinguish them from aspects that vary across particular cases.

In order to find non-contingent essences CAAP utilizes an abstracting methodology. For example, many arguments provided in the CAAP discourse treat people as isolated individuals whose mental operations are depicted as relatively linear and simplistic, sometimes almost like arithmetic equations. John Rawls provided one of the more famous depictions with his "veil of ignorance" where he asked that we consider people as if they, and we, were atomistic—without sex or gender, economic status, race, traditional or cultural background, and all of the other particularities which constitute who people are (Rawls, 1971). Relatedly, arguments in CAAP sometimes focus in on singular dimensions, and are developed therefrom. One understanding of moral motivation, for example, reads:

> When *P* judges that it would be morally right to φ, she is ordinarily motivated to φ; should *P* later become convinced that it would be wrong to φ and right to ψ instead, she ordinarily ceases to be motivated to φ and comes to be motivated to ψ (Smith, 1994, p. 61).

The complexities of social context, culture, personal history, biological influences, and interpersonal relationships are largely bracketed in order to isolate a specific philosophical question. This abstraction is often methodologically useful. It allows philosophers to investigate particular problems with considerable precision. Yet critics such as Walter Kaufmann (1978), Bernard Williams (1985), and Alasdair MacIntyre (1981) have argued that excessive abstraction risks distancing philosophical reflection from lived experience, social practices, and historical traditions. Whether or not one accepts such criticisms, they do highlight a persistent tension between the pursuit of universal principles or abstract concepts and attentiveness to particular contexts.

The methodological tendencies discussed here may also be productively related to the development of modern science. Ideals such as explicit representation, abstraction, formalization, and the isolations and control sought over variables are highly relevant in modern science, and are also important for the methodological tendencies of CAAP selected here. Although outside the scope of this article, research has been done indicating significant connections in these areas.[8]

[8] For discussion on the relationship between modern scientific ideals and the development of analytic philosophy in particular, see Rorty (1979) and Isaac (2019).

The purpose of these observations is not to criticize CAAP. Conceptual precision, rational argument, and abstraction have produced many of philosophy's most important achievements. Rather, the aim is simply to identify a methodological style that, as the following sections will suggest, bears an interesting structural resemblance to classical machine learning. Both seek clarity through explicit distinctions, rely heavily upon well-defined categories, and often operate most effectively in domains where variables can be isolated and carefully analyzed.

3 Supervised classical machine learning

Before deep learning (DL)—which underpins many of today's most impressive AI systems—earlier forms of AI relied much more heavily upon human-selected features and comparatively predetermined model structures. This earlier style of machine learning, which was especially influential from the mid-1990s through the early 2010s, is now often referred to as classical machine learning. While classical machine learning (CML) encompasses a diverse range of approaches, the present discussion focuses primarily on supervised learning systems that depend heavily upon hand-engineered features. So while CML itself is statistical and data-driven, we will mainly focus on its reliance on predefined categories, human-selected features, and predetermined model structures. These aspects provide the clearest contrast with contemporary deep learning systems.

Importantly, both CML and DL belong to a broader tradition of statistical modeling, and are both data-driven and pattern-sensitive. CML systems do learn correlations and other statistical patterns based on their training. The main contrast between CML and DL comes from the degree to which they rely on predetermined input and the way they organize data. Many CML systems depend much more on these than most DL systems. As will be discussed below, the latter are capable of learning multilayered representations from data, of iterative corrections, backpropagation, and other functions—which is why the difference in designations, i.e. CML versus DL, is broadly applicable.

In many respects, earlier forms of machine learning bear a structural resemblance to contemporary Anglophone academic philosophy (CAAP). Both rely heavily upon explicit categories, human-selected features, and comparatively predetermined structures in order to make sense of the world. Both function most effectively when working with relatively well-defined domains that can be analyzed through clear distinctions and controlled variables. Conversely, both encounter difficulties when faced with highly complex environments in which relevant patterns are difficult to isolate in advance. In this section we briefly outline several central features of supervised classical machine learning before turning to a comparison with CAAP.

Just as philosophy encompasses a wide range of approaches, so too does AI. It is therefore impossible to summarize all forms of machine learning in a short overview. For the purposes of this article, however, a broad account of supervised classical

machine learning is sufficient.[9] The goal is not to provide a comprehensive technical explanation, but rather to highlight several characteristics that are relevant to the comparison developed in later sections. In particular, we consider categories, feature engineering, interpretability, overfitting, and the problem of loopholes or shortcut solutions.[10]

In supervised machine learning, defining categories and engineering features often go hand in hand. Programmers first determine what they want the system to identify and then provide examples corresponding to those categories. Labels such as "spam," "fraud," or "cat" orient the model toward a particular prediction task. Engineers must then determine which characteristics of the data are likely to be relevant to that task.

Feature engineering refers to the process of identifying measurable characteristics of data that can be used for prediction. In many classical machine learning systems, these features must be specified by humans rather than extracted automatically from raw data. Engineers therefore rely upon domain knowledge to determine which aspects of the data are likely to be informative. The resulting model depends not only upon statistical optimization but, importantly, also upon prior human judgments regarding which features matter.

For example, if a classical machine learning system is designed to identify cats in images, engineers might specify features related to shapes, edges, textures, colors, or other measurable properties which are typically used to characterize cats. Metaphorically speaking, we might say that the model is given an algorithmic approximation of a "definition" of cat. The comparison should not be taken too literally, however. The system does not possess a philosophical definition of cat; rather, it relies upon engineered representations that function analogously by identifying characteristics expected to distinguish cats from other objects. The success of the model depends heavily upon the quality of these representations. They must be broad enough to capture natural variation while remaining narrow enough to avoid confusing cats with other animals.

Because these systems rely heavily upon human-engineered features, their effectiveness is often greatest in domains where relevant categories and characteristics are already reasonably well understood. In comparison with DL systems, these types of models are generally less capable of discovering entirely new features or identifying unexpected relationships within highly complex data. In other words, they are generally less capable of learning multilayered representations directly from raw data without prior relevant input. Their performance therefore depends substantially upon the adequacy of the conceptual structures supplied by programmers. Information that is not captured by those structures may remain inaccessible to the model.

Dependence on engineered features and predefined categories also contributes to a relatively high degree of interpretability. Compared with contemporary deep learning systems, many classical machine learning models allow researchers to trace how par-

[9] Other important machine-learning paradigms include unsupervised learning, semi-supervised learning, and reinforcement learning. A discussion of these approaches falls outside the scope of the present article.

[10] A related problem is "reward hacking," in which an AI system discovers strategies that maximize a specified reward function while violating the broader intention behind the task. For a discussion, see Skalse et al. (2022).

ticular inputs contribute to particular predictions. Definitions, features, weights, and outputs can often be examined individually. If a model performs poorly, the source of the problem can frequently be identified by examining the relevant features or modifying the way they are weighted. While not all classical machine learning systems are fully transparent, they are often considerably easier to interpret than contemporary deep learning models.

When such a model identifies a cat in an image, its prediction can often be reconstructed. The system analyzes relevant features, assigns them statistical weights, and generates an estimate regarding the likelihood that the image belongs to the category "cat." The process is comparatively transparent because the categories and features guiding the prediction have largely been specified in advance by human designers.

The same characteristics that contribute to interpretability can also contribute to overfitting. Google Developers define overfitting as "creating a model that matches (memorizing) the training set so closely that the model fails to make correct predictions on new data" (Google Developers, n.d.). In other words, an overfit model may perform extremely well on training data while struggling in new environments. The features that allow a model to identify patterns can become liabilities when they correspond too closely to accidental aspects of a particular dataset. The system may then treat random noise, bias, or coincidence as though it were genuinely predictive information.

Keeping with the image recognition example, imagine that most of the training images labeled "cat" happen to be brightly lit or taken in kitchens. The model may come to associate lighting conditions or kitchen features with catness. A small dog photographed in a brightly lit kitchen might then be incorrectly labeled a cat because the model has learned accidental correlations present in the training data. In such cases the model fails to generalize effectively beyond the circumstances under which it was trained; it cannot handle new complexity with a sufficient degree of flexibility.

Another limitation involves the problem of loopholes or shortcut solutions. Sometimes a model performs exactly as instructed while nevertheless violating the spirit of the task. This occurs because the model does not understand the meaning or purpose of its objective. Instead, it optimizes statistical relationships associated with the training data. If all images labeled "cat" contain a watermark, for example, the model may learn to identify the watermark rather than the animal itself. The resulting system appears successful during training while performing poorly when deployed elsewhere. Similar problems can emerge in far more consequential domains, including medicine, finance, and public policy—and these problems are not limited to CML.[11]

The discussion above is necessarily simplified, but it identifies several characteristics of supervised classical machine learning that will be important in what follows. In sum, the feature-engineered supervised CML systems considered here depend heavily upon predefined categories, engineered features, and explicit representations supplied by human designers. In the next section we shall argue that these character-

[11] Eliezer Yudkowsky, one of the most famous and insightful critics of artificial general intelligence, wrote about a "paperclip maximizer" to emphasize the dangers of loopholes. The example says that we might ask an AI to make paperclips, and then before we know it the whole world is turned into paperclips. Especially if AI becomes many more times "intelligent" (at least on some metrics) than humans, it will find all sorts of loopholes we could not even imagine (Yudkowsky, 2008).

istics bear an intriguing resemblance to several methodological tendencies identified earlier within contemporary Anglophone academic philosophy.

4 Classical machine learning (CML) and philosophy (CAAP)

One major theme of this article is the structural similarity between supervised classical machine learning (CML) and contemporary Anglophone academic philosophy (CAAP). There are numerous ways in which parallels between the two might be elucidated, but the discussion here focuses primarily on broad methodological resemblances. In particular, we highlight those features that can later be contrasted with deep learning. Accordingly, our concern is with the ways in which both CML and CAAP seek to make sense of the world through models that reduce complexity by relying upon explicit task categories, human-selected features, comparatively predetermined model structures, and relatively transparent forms of reasoning.[12]

The role played by definitions differs somewhat between CML and CAAP. In CAAP, conceptual clarification itself is often a central goal. This type of philosophical inquiry frequently seeks increasingly precise accounts of concepts and their relations to one another. CML, by contrast, does not develop definitions. Rather, it relies upon engineered representations and predefined features in order to function. (Strictly speaking, these are not "definitions" although they are metaphorically related.) Nevertheless, both feature-engineered CML systems and selected CAAP methodologies considered here exhibit a comparatively high level of dependence upon predetermined specified parameters, including features and model structures. The difference is that in CAAP conceptual precision is often the object of inquiry, whereas in CML it serves as a means for constructing effective predictive models.

Dependence upon explicit representations is closely related to a preference for transparent forms of reasoning. Scholars working within CAAP frequently develop rigorous arguments in support of conceptual distinctions or theoretical claims. Many aspire to forms of precision that, at least in some areas, approximate the standards associated with formal logic or mathematics. CML is itself mathematical in structure. Its operations can be described through statistical models, weighted variables, and clearly specified computational procedures. Although CML can be formalized more precisely than most philosophical arguments, both approaches share a commitment to making their reasoning explicit and traceable.

There are many benefits to systems that rely upon clearly specified parameters and logical rigor. Because their categories, objectives, and procedures are relatively stable, both CML and CAAP often possess a comparatively high degree of interpretability. In the case of CML, researchers can frequently identify which features contributed to a prediction and how they were weighted. In the case of CAAP, arguments can often be reconstructed and evaluated step by step. In many domains these charac-

[12] Again, as noted above this description of CML marks a difference with DL mainly as a matter of degree. Both CML and DL are statistical and data-driven, however, the degree to which they require, and the way they balance predefined structure and learned representations mark one significant difference between them.

teristics are substantial strengths. CML has been successfully applied to tasks such as fraud detection and housing-price prediction, while CAAP methods have contributed to the development of legal concepts, political institutions, and ethical frameworks. In these and many other contexts, both approaches have proven highly productive.

At the same time, the strengths of CML and CAAP can become limitations under certain conditions. In order to achieve clarity and precision, both often simplify phenomena by isolating particular variables or characteristics for analysis. This simplification can be methodologically valuable, but it may also make adaptation to more complex environments more difficult. A CML model trained to identify cats, for example, may struggle when presented with images that differ substantially from those found in its training data. Likewise, highly abstract philosophical models may encounter difficulties when applied to the complexity of lived experience, social practice, or historical context.

The point is not that either CML or CAAP is incapable of addressing complexity. Rather, both often approach complexity by reducing it to a set of manageable components that can be analyzed individually. This strategy can generate powerful insights, but it may also leave certain dimensions of a phenomenon underexplored. The resulting models are often most successful within the domains for which they were designed and may require significant modification when extended beyond those domains.

Philosophically, we might say that CML and CAAP struggle with relating to meaning and context in more concrete applications. Many CML systems can only make successful predictions for "cat" when enough weight of pre-given specified features is statistically high enough to meet the algorithm's threshold. When compared with deep learning models, CML is rather limited in many real-world arenas—its methods for grasping "cat" are limited by pregiven features. CAAP is often accused of struggling with meaning as well. While concepts might be extremely clear, they are often also sterile, and can struggle to resonate with non-experts. Explaining human motivation[13] according to succinct formulas, or arguing that it is primarily reason, or reason alone, and not emotions, social influence, tradition, culture, and biology which motivates people, does not meaningfully address the experience of this phenomenon for most people.[14] We may then note the selected approaches of CML and CAAP discussed here can both perform well in established frameworks, but may overlook important contextual features which fall outside of these frameworks.

These observations should not be understood as a rejection of either CML or CAAP. Both have produced important achievements and remain indispensable in many contexts. The aim is simply to identify a family of methodological similarities that will provide a useful contrast with deep learning. In the next section we turn to deep learning systems, which are very different in terms of how they tend to balance predefined structures and the representations they learn.

[13] Some critics have argued that philosophical discussions of agency and responsibility do not always engage extensively with empirical findings from neuroscience. Sapolsky (2023), for example, advances this criticism in his discussion of free will.

[14] Korsgaard's account of moral motivation, which emphasizes reflective endorsement and practical reason, provides a well-known example of an approach that assigns a central role to rational agency (Korsgaard, 1996).

5 The deep learning alternative

Nearly all of the AI that makes headlines, and indeed most of the AI we encounter in everyday life, is deep learning (DL) AI. Deep learning is so called because it relies on multiple layers of neural networks. In contrast to the comparatively straightforward nature of classical machine learning (CML), where there may be few or no intermediate layers between inputs and outputs, deep learning systems contain many layers through which data are continually transformed and reinterpreted. It is within these layers that much of their remarkable performance emerges. In what follows, we will highlight several contrasts between DL and CML that are particularly relevant to the concerns of this paper. Given the similarities identified between CML and CAAP in the previous sections, these contrasts are also suggestive of differences between DL and CAAP. We begin by considering the motivations behind deep learning.

The origins of DL computational structure can be traced to simplified models of biological neural networks (McCulloch & Pitts, 1943). Though contemporary DL systems are substantially different from actual brains, they were motivated by the observation that human intelligence appears capable of learning from experience without relying primarily upon explicit rules or definitions.[15] Unlike CML systems, which depend heavily upon human engineered features, DL systems attempt to learn effective representations directly from data (LeCun et al., 2015). Through exposure to large quantities[16] of training instances, they can gradually refine internal connections and representations, allowing them to come to recognize increasingly complex patterns. Although the mechanisms remain mathematical, the resulting capacities for pattern recognition, generalization, and adaptation can be meaningfully compared to certain aspects of human learning (Lipton, 2018). DL systems and human cognition are also similar in another respect: both are often difficult to explain in a fully transparent manner. While contemporary AI researchers continue to develop interpretability tools, the internal workings of deep neural networks remain substantially far more opaque than those of most CML models.

Interestingly, some of these contrasts between CML and DL are reminiscent of past philosophical debates concerning the nature of intelligence. Hubert Dreyfus famously criticized symbolic AI for assuming that intelligent behavior could be reduced to the manipulation of explicit rules and representations. Drawing upon phenomenology, Dreyfus argued that human expertise depends largely upon embodied, context-sensitive forms of coping that cannot be fully captured through formal rules (Dreyfus, 1992). Although Dreyfus wrote decades before the emergence of contemporary DL systems, the critique he developed is noteworthy because many of the limitations he found with symbolic approaches resemble difficulties encountered by CML systems that depend heavily upon predefined features and representations. DL AI does not vindicate Dreyfus's broader philosophical position, but it does provide a

[15] In special cases we can find exceptions, but on the whole this is not how humans are as a species. Language-learning provides a model, most (all?) people learn their native language through immersion, repetition, and feedback rather than through explicit instruction in grammatical rules or other formal principles.

[16] Some newer DL models can adapt to new tasks through transfer learning or few-shot prompting, although their underlying models may still have been trained with large data sets, see Brown (2020).

technological example of successful performance that relies less upon explicit symbolic specification and more upon patterns learned from experience.

Whereas CML depends heavily on predefined features and operates according to relatively straightforward statistical procedures, DL while still dependent on human-constructed architecture, objectives, and evaluation standards, is also able to learn useful features and representations through iterative exposure to data. It is not typically provided pre-defined features; it needs to find them itself. In other words, it learns which features are effective as part of its own training process. In this respect, CML relies more heavily upon explicit human instruction, whereas DL is capable of developing representations that were not directly specified by programmers.[17] Through continual refinement of internal connections and weights, DL systems become capable of identifying increasingly sophisticated patterns in data—and also in increasingly complex data.

To further illustrate this point, we can note that a DL system does not "see" an image of a cat as simply a collection of raw pixels, nor must it be explicitly instructed to search for whiskers, tails, or any particular shapes. Depending upon its exact architecture, the model can identify highly complex associations within the data and develop internal representations of cats that are nearly always far richer and more flexible than the engineered feature representations relied upon by CML systems.

Exactly what an internal representation of a cat is like in any DL system is difficult to determine. Through exposure to large numbers of training instances (which function like examples), the system develops multilayered internal representations that are highly effective but not fully transparent. And even with recent advances in explainable AI, researchers are often unable to specify precisely how a particular concept is encoded within a neural network. What is known is that DL systems can successfully identify, discuss, and even generate images of cats without relying upon explicit definitions or manually specified feature lists—which is basically how most humans encounter cats as well, whether images or actual.

The large majority of humans recognize, discuss, interact with, and can even draw cats without possessing a formal definition of "cat." Both the human brain and DL systems find successful performance emerges through repeated exposure to examples rather than through the application of explicitly articulated definitions. And as a result, both humans and the DL have representations of cats which are highly flexible, capable of generalizing to novel situations, and overall highly effective, even if their underlying processes or the representations themselves escape explicit description.

Much of the efficacy found in DL stems from its iterative learning process whereby it can locate and correct errors through backpropagation. Backpropagation, short for "backward propagation of errors," is a key component of many DL systems. Input data move through the layers of the network and generate a prediction. The model then compares this prediction against the desired output and calculates for any error. Backpropagation allows information about this error to move backwards through the network. The network then adjusts weights and connections so that future predictions can improve. In this sense, the system learns from its mistakes. Millions of such

[17] DL algorithms often start with weights which are randomly assigned and then. reconfigure them in processes of iterative exposure (LeCun et al., 2015).

adjustments can occur during training, allowing the network to develop increasingly effective representations in a relatively small amount of time.

The specific mathematical process by which weights are updated during backpropagation is typically gradient descent. Gradient descent iteratively adjusts weights and biases in directions that reduce prediction error, thereby minimizing the model's loss function (Goodfellow et al., 2016). Through repeated cycles of prediction, error correction, and parameter adjustment, a DL system gradually improves its performance on similar inputs. Repeated millions or even billions of times, this process allows the network to develop highly reliable representations of patterns such as catness.

Immersion in large quantities of data is essential to the successful functioning of DL. Compared with CML, which relies relatively more heavily upon engineered features and predetermined model structures, DL systems are highly dependent upon extensive training data. Without large numbers of training instances, DL models are unable to develop sophisticated internal representations. In this respect, DL depends upon data in a way that roughly parallels CML's dependence on engineered features. This also marks an important difference between contemporary AI systems and human learning, which often occurs with far fewer examples (i.e. training instances).

Compared with CML, DL is far more dynamic in the way that it develops representations for analyzing data. Its operations are highly non-linear, involve iterative learning processes (i.e. continual feedback and adjustment), and cannot be reduced into a small set of explicitly defined rules. DL processes convoluted data in a complex and holistic manner; the representations that emerge within these systems are relational,[18] distributed, and resistant to straightforward interpretation. DL systems are capable of identifying patterns and relationships that would be difficult or impossible to specify manually through feature engineering alone. They can pick up on elusive correlations that are inaccessible to CML models, and can be puzzling even to their programmers.

The emergence of generative AI has made these characteristics especially visible. Large language models (LLMs), for example, are trained not through explicit definitions of concepts, grammatical rules, or symbolic representations of meaning, but through exposure to vast amounts of linguistic data.[19] Through this process they develop internal representations that allow them to generate coherent text, respond to questions, translate languages, summarize arguments, and perform many other tasks that were once thought to require explicit conceptual understanding. Whether such systems genuinely understand remains a matter of philosophical debate.[20] What is significant for present purposes is that their success often depends less upon manually specified rules and definitions than upon the emergence of effective patterns through

[18] Here "relational" refers to statistical dependencies, and should not be confused with human relationships, which will be discussed below.

[19] Murray Shanahan expresses a similar point when he argues that contemporary LLMs are generative models trained on vast corpora of human language, and not systems built around explicitly encoded conceptual structures.

(Shanahan, 2024).

[20] For a survey of this debate, see Mitchell and Krakauer (2023). They look at competing arguments concerning whether contemporary large language models understand language in any meaningful sense, and what it might mean to attribute understanding to such systems.

large-scale exposure to training instances (functioning as examples in some sense). In this respect, generative AI further illustrates the contrast between approaches that rely primarily upon explicit representation and those that learn through iterative engagement with experience.

In some respects,[21] deep learning resembles the way native speakers acquire language. They do not begin with formal definitions or explicit grammatical rules—in fact they only learn these through difficult processes. It is rather through immersion, repetition, feedback, and continual adjustment, they gradually develop increasingly sophisticated linguistic competencies. In both cases, learning is driven primarily by exposure to examples and ongoing correction rather than by the direct transmission of explicit definitions. Much of the resulting competence emerges implicitly through experience and resists complete formal articulation.[22] While many philosophical traditions emphasize analogous forms of learning, the next section focuses on early Confucian and Daoist approaches, both of which place considerable emphasis on examples, cultivation, and practical engagement rather than abstract definition.

6 Selected approaches in early Chinese philosophy

Critically, some approaches found in early Confucian and Daoist philosophy differ in important respects from the characteristics of contemporary Anglophone academic philosophy (CAAP) outlined above.[23] Just as classical machine learning (CML) and deep learning (DL) are both forms of artificial intelligence yet operate according to different structures and learning processes, various methods found in early Confucian and Daoist philosophies (CDP) and certain ones in CAAP proffer ways of philosophical reflection that can be distinguished from one another. On each of the major characteristics identified above—namely, the prioritization of reason, conceptual clarification, and the search for essential features—we can find alternative orientations in selected early Confucian and Daoist texts. The discussion that follows is necessarily selective and does not attempt to characterize all Chinese philosophy, or the span of Confucian and Daoist traditions. Rather, it focuses primarily on certain themes found in early Confucian and Daoist texts, particularly those which lend themselves to be distinguished from CAAP and also compared with DL.

[21] There are, of course, major differences as well. In a general sense, for example, human learning is embodied and socially situated, whereas DL learns through statistical optimization.

[22] Dreyfus (1972) drew a similar conclusion in his critique of symbolic AI, arguing that intelligent performance often depends upon forms of practical know-how and situational responsiveness that cannot be fully captured through explicit rules.

[23] These approaches are developed in specific historical and social settings, they are not timeless characteristics of "Chinese thought," and alternative methods, especially in Mohist and Legalist texts, are also part of the Chinese tradition. For rich accounts of the historical and social contexts these ideas were cultivated in see, for example, Graham (1989) or Nylan (2001).

Reason and logical argument are hallmarks of CAAP. Early Chinese philosophy also contains rational reflection and argumentation.[24] Importantly, in many influential texts, emotions and reasoning are not sharply separated—and experience, personal cultivation, and particular relationships and circumstances are all taken as thoroughly intertwined with emotions and reason as well.[25] The early concept of *xin* 心 is frequently translated as "heart-mind" precisely because it encompasses both cognitive and affective dimensions. Thought and feeling are not treated as faculties which can be meaningfully distinguished from one another, and cultivation of one nearly always implies cultivation of the other. Li Zehou (d. 2021), for example, characterizes the human person as constituted by emotional-rational structures rather than by sharply separated domains of reason and emotion. As a result, much of the argumentation found in early Chinese philosophy appeals simultaneously to reflection, feeling, experience, and the particular situation at hand.

Roger Ames has similarly argued that Chinese thought is best understood as relational, (Ames, 2011) and, in their collaborative work, Ames and Henry Rosemont have been particularly critical of interpretations that associate Chinese thought with isolated rational subjects (Ames & Rosemont, 2009). On their interpretation, persons are constituted through relationships and roles, and ethical reflection concerns the cultivation of those relationships rather than the analysis or application of abstract principles. Such approaches stand in noticeable contrast to philosophical traditions that seek to isolate reasoning from broader social and emotional contexts.

Unsurprisingly, the way concepts function is different as well. Some of the most important concepts in early Chinese thought, including *dao* 道 (way, path, method) and *xiao* 孝 (filial piety), resist fixed definition. Working specifically on the function of concepts in Classical Chinese philosophy, Chad Hansen argues that these concepts function less as abstract definitions and more as practical guides to action (Hansen, 1992).[26] Rather than seeking necessary and sufficient conditions, the emphasis is more often on appropriateness, cultivation, and contextual realizing or embodying. These all require proper attunement with particular contexts, including, of course, relations with others.

Confucius' discussions in the *Analects* provide illustrative examples. For example, when asked by his students "what is *xiao* 孝?" (*xiao* being perhaps the central value in Confucianism) he repeatedly answers through examples and context-sensitive

[24] It is worth reminding that we are using an extremely limited scope in discussing Chinese philosophy—we are referring almost exclusively to Pre-Qin Confucian and Daoist texts, as well as certain lineages of their development. In Legalist and Mohist texts we do find many exceptions to what is said here. Especially what is said below in terms of arguments, linearity, and logic. Mohist texts represent a huge exception as well.

[25] Some readers may object to this characterization. Nevertheless, many scholars have argued that important strands of Chinese philosophy employ styles of reasoning and argumentation that differ in significant respects from those characteristics of contemporary Anglophone philosophy or classical Indian philosophical traditions.

[26] Relatedly, early Chinese lexicographical works such as the *Shuowen Jiezi* 說文解字 (*Explaining Simple and Analyzing Compound Characters*) often explain characters through associations, component analysis, examples, and patterns of usage rather than through the kind of strict necessary-and-sufficient definitions familiar from many modern dictionaries (Xu, 1963).

responses rather than through a single universal definition.[27] The students appear to seek a general account, but Confucius consistently redirects attention toward concrete situations and particular relationships. When we compare to Plato's writings, we find the roles of philosopher/master and interlocutor reversed. Socrates is the one asking for the universal definition while those he speaks with provide examples.

The comparison with Plato's writings also serves us well when looking at CAAP's attempts to distinguish the essential from non-essential. This is a chief concern of Socrates. It is why he wants to get away from particular examples and see what they share in common. Confucius frequently advocates thinking in the opposite orientation— seeking to gather more details about the particulars in order to better understand the entire situation, including the people involved. In a most telling exchange in the *Analects*, when someone says that in considering a person of high moral cultivation only essential aspects matter and the ornamental do not, he is fervently rebuffed. (12.8) And in Daoist philosophy, where *dao* is most important, it is discussed as entirely ineffable (*Laozi* 1), and as the source of all things, including concrete particulars (*Laozi* 42). In other words, attempts to sharply separate the essential from the non-essential are not a central philosophical task in CDP.

The framework for philosophical reflection in certain areas of early Chinese thought is therefore different in several important respects. Here we will outline a few key characteristics, starting with the concern involving establishing appropriate correspondence between names and actualities. Names refer not only to objects but also to roles, values, relationships, and forms of conduct. Actualities refer to the concrete situations, persons, and circumstances to which such names apply. Much of Chinese philosophy can be understood as an attempt to negotiate the relationship between the two, although Daoist thinkers differ by arguing they can never really match.

Since concepts are not defined through strict necessary-and-sufficient conditions, people have only a vague understanding of what it means to be, for example, a good sibling, a sincere teacher, or a virtuous ruler. Yet these roles must still be enacted. Understanding develops through experience, participation, reflection, and cultivation rather than through the acquisition of precise definitions. In this respect, actually being efficacious in interacting with others is the highest standard.

Even while concepts themselves are not rigidly defined, the tradition is rich in examples, analogies, historical figures, and exemplary models. Such examples function as guides for action rather than as abstract theories—they are suggestions for how one might feel when they embody whatever name (virtue/value/role). And these examples can be found in real life but also in canonical texts (*jing* 經). They must then be applied through processes of contextual judgment (*quan* 權). This *jing-quan* dynamic can be understood as "balancing guidelines with particulars." What matters most is the ability to respond appropriately within concrete situations.

At first, this process may involve a good deal of rote learning and mechanical application—especially in Confucian contexts. Students memorize classics, examples, rituals, and established practices and use them to navigate their lives. Over time, however, these practices are expected to become deeply embodied. Through continued participation, individuals learn not merely what to do, but how to perceive situ-

[27] Sometimes Confucius' responses are even contradictory (Chen, 2009).

ations differently. In other words, their tacit knowledge becomes cultivated through the imitation of what they have memorized. Zhu Xi (d. 1200) refers to this process as *tihui* 體會, which means "to realize," "learn from experience," and "embody." In Confucian thought more generally, this process is often described as cultivation (xiu 修).

The *dao* or "way" that various schools of thought in the Chinese tradition promote thus refers not to something static and outside of human behavior, it is made up (at least in part[28]) by what people do. Especially in terms of the socio-political and ethical dimensions, people do not simply align with *dao*, they broaden it (*Analects* 15.29), it is through human action that it is constantly made and re-made (*Zhuangzi* 2.6). The classics, examples, and doctrines, one uses to reflect on themselves and the world are models that came from humans, and must constantly be re-established. People learn about the way to cultivate themselves, and in doing so also cultivate the way.[29]

7 Deep learning and selected Confucian and Daoist approaches

This comparison between the selected approaches of Confucian and Daoist philosophies (CDP) outlined above and deep learning (DL) systems is not a claim that either has any direct relation to one another. Rather, we hope to develop the suggestion that both exhibit a similar methodological orientation, and can deal with similar problems in effective ways. In each case, successful performance emerges less through explicit definitions and formal rules than through sustained engagement with examples, patterns, models, and contextual responsiveness. The recent success of deep learning therefore provides a contemporary technological illustration of a broader philosophical possibility: that understanding need not always depend upon complete explicit articulation, and that analysis of concepts does not necessarily lead to better ways of understanding or interacting in the world. Recent developments in AI invite renewed reflection on whether examples, cultivation, tacit knowledge, and context-sensitive judgment might play a larger role in philosophical inquiry than is often acknowledged. And it is in this sense that we are arguing contemporary that Anglophone academic (CAAP) stands to learn from both DL and CDP.

To begin, we can note that DL learns from training instances and emphasis in the selected Confucian and Daoist approaches is on significant examples, models, guidelines, and exemplars. Though the similarity is limited, both DL and CDP do share in

[28] In some contexts *dao* might also be used to signal something cosmological, nevertheless, it is rarely, if ever, divorced from humans or their behaviors.

[29] We may further note that early Chinese thinkers did not assume the world, people, or interactions, could be entirely known. Only some dimensions were accessible to humans, and relying purely on reason would narrow the scope of what could be comprehended even more. It was not much before 1900 that the word "logic" (*luoji* 邏輯) was introduced into Chinese. Before then speaking about *daoli* 道理 was used to express "accord with *dao*" which is functionally loosely relatable to "that makes sense," or "being reasonable." In certain contexts, *daoli* expresses ethical order, or even ethical doctrine—perhaps best understood as "an appropriate way." It is still something to be "cultivated" unlike an abstract and universal truth or reason and logic.

that patterns of effective responsiveness are developed through repeated encounters with particulars. Compared with certain models of CML and methods of CAAP, the selection of DL and CDP considered here rely less on pregiven specificity. In other words, DL and CDP emphasize engagement with particulars in learning, including various forms of examples (or training instances), patterns, and "experience." In this way we can say that they develop through resonating responsiveness to the world—more concretely, their competence and "understanding" emerge gradually through iterative engagement with concrete situations. Neither is successful because of the application of explicit definitions or universal rules. For DL systems, repeated exposure to data results in increasingly effective representations (or "understanding"), while CDP traditions emphasize cultivation through participation, reflection, imitation, and practice. They are, of course, different, but the general orientation of respective approaches overlaps broadly in suggesting that effective responsiveness can emerge from sustained engagement with particulars rather than from possession of exhaustive concepts or formulations.

Compared with contemporary Anglophone academic philosophy (CAAP) and classical machine learning (CML), both DL and CDP place less emphasis on explicit definitions and predetermined procedures. As discussed above, neither depends primarily upon fixed conceptual boundaries in order to function effectively. Instead, both rely upon the development of flexible patterns of responsiveness in interacting with particulars and engaging in iterative processes. In the case of CDP, rituals, virtues, exemplary figures, and canonical texts provide important forms of guidance, but they are not ordinarily treated as rigid rules to be mechanically applied. (The most rigid guidelines we can find in CDP are the rituals which Confucianism expounds, but even these are subject to constant adjustment through consideration of situational factors.) Through the *jing-quan* dynamic, inherited models must continually be interpreted and adjusted in light of particular circumstances. Values and virtues are seen as evolving patterns of efficacious conduct that are refined through constant practice and continual attunement. The goal is therefore not strict conformity to abstract prescriptions—although this may be useful at times as well—but more importantly, the cultivation sought in CDP is the kind which embodies an ability to respond appropriately within changing contexts.

DL systems exhibit structures which are broadly analogous to CDP. Engineers do program architectures, training procedures, and optimization objectives, however, the representations that emerge during training are not the result of human specification. Internal representations emerge from repeated exposure to data. So the effectiveness DL systems display in tasks such as recognition, prediction, and generation (including LLMs) stems largely from this capacity to learn patterns that would be difficult or impossible to specify explicitly through feature engineering alone. Through millions of iterations, DL neural networks continually adjust their parameters in response to feedback, and as a result they can come up with increasingly sophisticated ways of responding to the data with which they interact.

Part of the reason both DL and CDP can function effectively without exhaustive definitions stems from their respective iterative learning processes. In Confucian and Daoist traditions, people learn from real-life or past models and examples, or through rituals, etiquette and other effective patterns of conduct. People may start

off by mechanically following what they have learned, but they are supposed to constantly test it, and adapt behaviors according to the specifics of the situation. Through continual reflection, participation, and adaptation, what begins as external guidance gradually becomes embodied competence—an understanding which can never be made fully explicit. Established teachings serve as starting points to guide people along a general direction, and they are also important points to reflect on along the way, but that is it. The goal is cultivation of an ability to respond appropriately to changing circumstances, and this can only ever be lived.

While the abovementioned discussion covers the Confucian notion of cultivation particularly well, in broad ways it is also related to the discussions of *wuwei* 無為 and *ziran* 自然 prioritized in Daoist texts. *Wuwei* can be understood as "non-action" in the sense of not acting in a way that is distanced from circumstances or overly attached to prior conceptions. Relatedly, *ziran* or "self-so" calls attention to interacting through moving with the particulars of oneself, others, and the environment. Both are highly resistant to explicit explanations and call for high levels of attunement and constant context-sensitive readjustments.

DL is also an iterative process where the system learns from mistakes. Through backpropagation and gradient descent, models continually modify their internal parameters in response to prediction errors. Learning and "understanding" occur through repeated cycles of prediction, feedback, and refinement. The resulting representations or "understanding" are not directly encoded by programmers but emerge through the model's ongoing engagement with data. DL and CDP both involve processes of attunement where efficacy results from dynamic adaptation to experience, and not from the application of complete prior specifications.[30]

In sum, we can note that CDP and DL both highlight forms of efficacy and "understanding" which emerge primarily through iterative engagement with experience. Their particular effectiveness derives not primarily from the rote application of explicit rules or narrowly defined concepts, but from the development of flexible patterns of responsiveness. Again, this comparison is not intended to collapse the distinction between philosophy and technology. Rather, we are arguing that recent success of deep learning provides a striking contemporary example of how learning through examples, patterns, and contextual responsiveness can generate forms of understanding and effectiveness that are difficult to achieve through explicit definitions alone.

8 Conclusion: What philosophy should learn from AI

The success of contemporary deep learning and generative AI provides a reason to reconsider philosophical methodologies that privilege explicit definition, conceptual analysis, formal explanation, and predetermined structures. Specifically, in this paper

[30] Although outside the scope of this article, Polanyi's notion of tacit knowledge could provide a useful framework for understanding this overlap. Just as skilled practitioners often know more than they can explicitly articulate, both DL systems and CDP involve types of efficacy which exceed straightforward description. However, human tacit knowledge is embodied, socially situated, and exists in actual human communities, which is quite different from DL.

we have sought to develop this suggestion through a series of structural comparisons, which themselves illustrate the kind of pattern-based methodology under discussion. We have argued that certain methodologies prominent in contemporary Anglophone academic philosophy (CAAP) and classical machine learning (CML) both tend to rely heavily upon explicit definitions, predetermined structures, and relatively transparent forms of procedural reasoning. By contrast,[31] we can understand selected approaches in deep learning (DL) and the Confucian and Daoist philosophies (CDP) as placing greater emphasis upon examples, patterns, cultivation, and context-sensitive responsiveness.

DL does not prove the validity or correctness of CDP any more than CML evidences the effectiveness of CAAP. Our argument is that CAAP could incorporate some of the methodologies exhibited by DL systems, especially those which are broadly similar to what is found in CDP. This would mean philosophical inquiry would also look at concrete examples and patterns more seriously, that cultivation and historical and social situatedness would become influential, and that learned context-sensitive responsiveness would become an ideal. This is not to say that we think conceptual analysis should be abandoned, nor that CAAP is somehow mistaken in its analysis of concepts aimed at clarity and precision. Like CML, such an approach remains extremely valuable in many contexts. What we do want to suggest is that this represents only one family of intellectual tools among many.

For example, many approaches to philosophical inquiry today are characterized by abstract reasoning, concept analysis, and theoretical discussions. Here hypothetical questions and responses are generally taken as legitimate ways to expound on problems or prove a point. However, from the perspective of DL and CDP these would be understood as pregiven structures or models, which we would want to adjust through iterative processes relying on concrete experience and actual cases. By no means does this imply that hypothetical questions, theoretical arguments, concept analysis, or abstract reasoning should be dismissed or taken as less important. It does, however, mean we can make more space for methods which incorporate feedback from actual practice.

Accordingly, philosophers would have new resources for thinking about "error correction." Starting with a variety of possibilities, such as different examples, models, doctrines, or provisional concepts, they could examine how they function in concrete circumstances, testing patterns against a variety of cases and criticisms, and in terms of practical consequences and actual experiences, and then revise them iteratively. In this way error correction would not be comprised of theoretical clarification of a unified objective, but an ongoing human process in which the problem, its relevance, and even the standards of success would be subject to continual updates.

Philosophy should not imitate artificial intelligence, but it can learn from it. The recent success of DL provides a salient counterexample to an assumption that robust competence must be preceded by conceptual decomposition. A space is thereby is

[31] The divisions we draw between the similarities highlighted in CAAP/CML and DL/CDP are just one way to understand their complex relationships. CML and DL are similar in being data-driven and statistical, and another way to see their difference is through how they balance their predefined structures with data. Likewise, the selected approaches of CAAP and CDP are just particular aspects of traditions which are complex, socially and historically situated, and change over time.

opened wherein philosophers may learn from other approaches—including CDP—and potentially broaden the range of methods through which inquiry can begin, develop, and correct itself.

Acknowledgements None.

Funding This article is supported by "Supported by the Fundamental Research Funds for the Central Universities".

Data availability None.

Declarations

Competing interests None.

References

American Philosophical Association (n.d.). *Philosophy: A brief guide for undergraduates*. https://www.apaonline.org/page/undergraduates

Ames, R. T. (2011). *Confucian role ethics: A vocabulary*. University of Hawai'i.

Ames, R. T., & Rosemont, H. Jr. (2009). *The Chinese classic of family reverence: A philosophical translation of the Xiaojing*. University of Hawai'i.

Brown, T. B., et al. (2020). *Language models are few-shot learners*. NeurIPS.

Chen, L. 陳來. (2009). *Gudai sixiang wenhua de shijie* 古代思想文化的世界; The world of ancient thought and culture. SDX Joint Publishing Company (生活·读书·新知三联书店).

Dreyfus, H. L. (1972). *What computers can't do: A critique of artificial reason*. Harper and Row.

Dreyfus, H. L. (1992). *What computers still can't do: A critique of artificial reason*. MIT Press.

Goodfellow, I., Bengio, Y., & Courville, A. (2016). *Deep learning*. MIT Press.

Google Developers (n.d.). *Overfitting*. Machine Learning Crash Course. Retrieved from https://developers.google.com/machine-learning/crash-course/overfitting/overfitting

Graham, A. C. (1989). *Disputers of the Tao: Philosophical argument in ancient China*. Open Court.

Hansen, C. (1992). *A Daoist theory of Chinese thought: A philosophical interpretation*. Oxford University Press.

Isaac, J. (2019). The many faces of analytic philosophy. In *the Cambridge history of modern European thought* (Vol. 2, 176–199).

Kaufmann, W. (1978). *From Shakespeare to existentialism* (expanded ed.). Princeton University Press.

Korsgaard, C. M. (1996). *The sources of normativity*. Cambridge University Press.

LeCun, Y., Bengio, Y., & Hinton, G. (2015). Deep learning. *Nature*, *521*(7553), 436–444.

Lipton, Z. C. (2018). The mythos of model interpretability. *Queue*, *16*(3), 31–57.

MacIntyre, A. (1981). *After virtue*. University of Notre Dame.

McCulloch, W. S., & Pitts, W. (1943). A logical calculus of the ideas immanent in nervous activity. *Bulletin of Mathematical Biophysics*, *5*(4), 115–133.

Mitchell, M., & Krakauer, D. C. (2023). The debate over understanding in AI's large language models. *Proceedings of the National Academy of Sciences*, *120*(13).

Moran, D. (2000). *Introduction to phenomenology*. Routledge.

Nagel, T. (1979). *Mortal questions*. Cambridge University Press.

Nylan, M. (2001). *The five "Confucian" classics*. Yale University Press.

Plato (1997). Meno (G. M. A. Grube, Trans.; J. M. Cooper, Rev.). In J. M. Cooper (Ed.), Plato: Complete works (pp. 870–897). Hackett.

Rawls, J. (1971). *A theory of justice*. Harvard University Press.

Rolffs, M. (2023). Resultant moral luck and the scope of moral responsibility. *Philosophical Studies*, *180*(8), 2355–2376. https://doi.org/10.1007/s11098-023-01994-8

Rorty, R. (1979). *Philosophy and the mirror of nature*. Princeton University Press.

Sapolsky, R. M. (2023). *Determined: A science of life without free will*. Penguin.
Shanahan, M. (2024). Talking about large language models. *Communications of the ACM*, *67*(2), 68–79.
Skalse, J., Howe, N. H. R., Krasheninnikov, D., & Krueger, D. (2022). Defining and characterizing reward hacking. *arXiv* preprint arXiv:2209.13085.
Smith, M. (1994). *The moral problem*. Blackwell.
Williams, B. (1985). *Ethics and the limits of philosophy*. Harvard University Press.
Wittgenstein, L. (2009). *Philosophical investigations*. Wiley-Blackwell.
Xu, & Shen 許慎. (1963). *Shuowen jiezi* 說文解字 [Explaining simple and analyzing compound characters]. Zhonghua Shuju 中華書局.
Yudkowsky, E. (2008). Artificial intelligence as a positive and negative factor in global risk. In N. Bostrom, & M. Cirkovic (Eds.), *Global catastrophic risks* (pp. 308–345). Oxford University Press.

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.

Authors and Affiliations

Paul J. D'Ambrosio[1,2]

✉ Paul J. D'Ambrosio
pauljdambrosio@hotmail.com

[1] Institute of Modern Chinese Thought and Culture, Philosophy Department, East China Normal University, Shanghai, People's Republic of China

[2] Ma Yifu Academy, Zhejiang University, Hangzhou, People's Republic of China